

Deinterleaving Mixed Radar Pulse Sequences in Complex Electromagnetic Environments

JINSHUO HOU[✉]

University of Chinese Academy of Sciences, Beijing, China

QUNYING ZHANG[✉]

Chinese Academy of Sciences Aerospace Information Research Institute, Beijing, China

PENG ZHANG[✉]

Beijing Institute of Tracking and Telecommunication Technology, Beijing, China

WEI LU

XIAOJUN LIU[✉], Member, IEEE

GUANGYOU FANG[✉]

Chinese Academy of Sciences Aerospace Information Research Institute, Beijing, China

Abstract— With the increasing number of emitters and interference, coupled with the development of low probability of intercept techniques, the complexity of operational systems, and the integrated application of anti-jamming technologies, the pulse stream intercepted by reconnaissance receivers in modern electronic warfare exhibits characteristics such as high pulse missing rate, numerous spurious pulses, large observation noise, and complex pulse repetition interval (PRI) modulations. The previously described pulse stream is referred to as mixed radar pulse sequences in this article. Deinterleaving mixed radar pulse sequences is highly challenging due to the complexity of the electromagnetic environment. The existing methods suffer from issues such as insufficient consideration of complex electromagnetic environments and high computational complexity. This article introduces the

concept of the PRI spectrum and analyzes its properties. Based on this representation, a PRI estimation algorithm that adapts to different PRI types is proposed. Additionally, an improvement to the pulse sequence retrieval algorithm based on directed acyclic graph (DAG) is proposed, enabling accurate separation of different pulse sequences. Simulation results show that the algorithm can successfully deinterleave of the mixed pulse sequences. When the jitter bound is 15%, the miss rate is 30%, and the spurious pulse count is 200 within 60 milliseconds, merr and PRism achieved 0.71% and 80.79%, respectively, demonstrating strong performance.

Index Terms— PRI estimation, pulse deinterleaving, radar signal, sequence retrieval

I. INTRODUCTION

Electronic warfare is a pivotal component of both offensive and defensive strategies in contemporary warfare. Due to the widespread use of radar in military applications, radar countermeasures have always played a dominant role in electronic warfare. The intercepted signal in radar countermeasures is typically a randomly interleaved signal stream formed by multiple radar pulse sequences, and is also accompanied by pulse arrival time (TOA) measurement noise, pulse missing, and spurious pulses. In this article, this is referred to as mixed radar pulse sequences. Only by separating the signals from different radars in the dense mixed radar pulse sequence can the attributes and threat levels of the emitters be identified, enabling appropriate countermeasures.

The radar signal deinterleaving process typically involves two stages: predeinterleaving and main deinterleaving. In the predeinterleaving stage, radar pulse sequences are clustered based on features such as angle of arrival, carrier frequency, pulse width, pulse amplitude, dispersing the pulse stream to prepare it for the subsequent main deinterleaving process. In the main deinterleaving stage, the TOA is utilized to estimate the pulse repetition interval (PRI), further completing the signal deinterleaving process.

As radar technology rapidly advances, the PRI modulation of signals intercepted by reconnaissance receivers has become increasingly complex. In addition to the fixed PRI widely used in conventional radars, there are also multiple other PRI modulation techniques, such as jitter PRI, staggered PRI, and sliding PRI. Furthermore, with the increasingly complex electromagnetic environment and advances in anti-jamming technologies, mixed radar pulse sequences are affected not only by pulse missing and spurious pulses but also by noise in TOA measurements. These factors make the deinterleaving of mixed radar pulse sequences highly challenging. This work primarily focuses on separating interleaved radar pulse sequences under conditions involving of TOA noise, missing pulses, and spurious pulses.

An early sequence search algorithm for radar signal deinterleaving using only TOA would attempt to match sequences in the sample by considering all possible PRIs and phases [1]. However, the true PRIs and phases correspond to only a fraction of these possibilities. Davies

Manuscript received XXXXX 00, 0000; revised XXXXX 00, 0000; accepted XXXXX 00, 0000.

Authors' addresses: Jinshuo Hou is with the Key Laboratory of Electromagnetic Radiation and Sensing Technology, Chinese Academy of Sciences Aerospace Information Research Institute, Beijing, 100190, China, and also with University of the Chinese Academy of Sciences School of Electronic Electrical and Communication Engineering, Beijing, 100190, China (e-mail: houjinshuo23@mails.ucas.ac.cn). Qunying Zhang, Wei Lu, Xiaojun Liu, and Guangyou Fang are with the Key Laboratory of Electromagnetic Radiation and Sensing Technology, Chinese Academy of Sciences Aerospace Information Research Institute, Beijing, 100190, China (e-mail: qyzhang@mail.ie.ac.cn, luwei@aircas.ac.cn, lxjdr@mail.ie.ac.cn, gyfang@mail.ie.ac.cn). Peng Zhang is with Beijing Institute of Tracking and Telecommunication Technology, Beijing 100094, China (e-mail: zhangpengbitt@163.com). (Corresponding author: Qunying Zhang and Peng Zhang.)

0018-9251 © 2022 IEEE

and Hollands [2] constrained the search to actual values through the selection of a neighboring pulse pair. Since then, many algorithms have been proposed to improve deinterleaving efficiency and accuracy.

The simplest algorithm based on TOA intervals is the TOA difference histogram. In 1989, Mardia proposed the CDIF algorithm [1] to determine possible PRI values, further improving search efficiency and accuracy. However, CDIF is highly sensitive to PRI jitter, and even in the case of very simple pulse sequences, it requires multi-level differencing. To address this, Milojevic [3] proposed the SDIF algorithm. The SDIF algorithm discards the cumulative histogram and second harmonic conditions of CDIF, introducing a subharmonic detection algorithm to reduce computational load and improve adaptability to missing pulses. However, it still remains sensitive to PRI jitter. Liu [4] proposed an enhanced SDIF algorithm using clustering algorithms and PRI transformation. This enhanced algorithm exhibits greater robustness against PRI jitter, but it's not applicable to different PRI types. In addition to the TOA difference histogram, some researchers use the TOA difference matrix to identify fixed PRI pulse sequences [5]. The PRI transformation algorithm [6] determines PRI values by calculating the autocorrelation integral. It effectively suppresses the harmonics, but is sensitive to PRI jitter. NISHIGUCHI [7] made two improvements to the PRI transformation algorithm: time reference adjustment and PRI bin overlap. The improved PRI transformation algorithm demonstrates better robustness against PRI jitter, but it is not adaptable to various PRI types.

To better estimate the potential PRI in mixed pulse sequences, period estimation methods have naturally been applied to radar signal deinterleaving, as determining the PRI value is essentially a form of periodic estimation. In 1996, Orsi et al. [8] applied the discrete Fourier transform and analyzed interleaved pulse trains in the frequency domain to estimate both the number of pulses and their periods in a sequence, but this method is not suitable for different PRI types. Sethares [9] proposed a method to determine the signal period by projecting the pulse sequence onto different periodic subspaces. However, due to the lack of orthogonality among periodic subspaces, the projection energy depends on the order of projection in the subspaces. To address this issue, Muresan [10] proposed an exact periodic subspace (EPS) algorithm, which generates orthogonal periodic subspaces to eliminate the non-orthogonality problem. However, this orthogonality condition is no longer valid when the signal duration is constrained. In 2017, Pei [11] introduced a closed-form orthogonal Ramanujan integer basis. Its orthogonality among periodic subspaces is independent of the pulse sequence length. Leveraging this foundation, a correlation matching method (CMM) [12] was developed to estimate the potential PRI values. This algorithm demonstrates robust adaptability to missing pulses, but its performance is affected by PRI jitter.

Other time-domain methods have also been applied to radar signal deinterleaving. Moore et al. [13] formulated the pulse train deinterleaving problem as a stochastic discrete-time dynamic linear model (DLM). They used the Kalman filter [14] and extended Kalman [15] filter to implement pulse sequence deinterleaving. However, this approach requires prior knowledge of the number of sources. Logothetis et al. [16] proposed an interval-amplitude algorithm for separating stochastic pulse trains. However, similar to Moore's method, it requires prior knowledge of the number of sources. Liu et al. [17][18] compared the pulse deinterleaving problem to a multi-target tracking system, using techniques from multi-target tracking to solve the pulse deinterleaving problem. However, it is sensitive to PRI jitter. Xie et al. [19] investigated the pulse sequence retrieval problem, focusing on cases where the PRI is known. The approach reformulates pulse deinterleaving as a longest path search problem in a DAG, but it is intended for fixed sequences and is not applicable to different PRI types.

The previously mentioned works do not provide solutions for deinterleaving mixed radar sequences in the presence of missing and spurious pulses. Cheng [20] proposed a method that combines enhanced CMM and improved PRI transformation, achieving effective deinterleaving of mixed radar sequences. However, it overlooks the challenges posed by spurious pulses. Furthermore, for a staggered sequence, the algorithm can deinterleave only one subsequence per iteration, leading to high computational complexity.

In recent years, the advent of the third wave of artificial intelligence has led to the application of deep learning techniques in radar pulse deinterleaving [21][22][23][24]. Although these methods perform well, they generally require large amounts of data for training and have limited real-time capabilities, posing challenges in practical applications. Therefore, further research into traditional radar signal deinterleaving algorithms is required.

To address the aforementioned limitations, this article proposes a novel algorithm for deinterleaving mixed pulse sequences. The contributions of our work can be summarized as follows.

1) A robust PRI estimation method leveraging the PRI spectrum is introduced. Based on a theoretical analysis of the statistical characteristics of the PRI spectrum, we demonstrate that the proposed method is robust against TOA noise, missing pulses, and spurious pulses, and that it can adapt to various PRI modulation types. Motivated by the insights gained from this analysis, a segmented PRI spectrum is further employed to address the effects of PRI jitter. Moreover, the phase information of the PRI spectrum is exploited to estimate sub-PRI in complex PRI-modulated sequences.

2) An enhanced DAG-based pulse sequence retrieval algorithm is developed. In contrast to the traditional DAG-based method, which is typically confined to fixed or jitter PRI types, the proposed algorithm generalizes to a wider range of PRI modulations. Additionally, for complex

PRI patterns, it can extract all relevant sub-sequences in a single iteration, significantly reducing the overall computational cost.

The remainder of this paper is organized as follows. Section II provides an introduction to the TOA progression for mixed radar pulse sequences and details each component of the proposed algorithm. In Section III, we conduct simulation experiments to validate the performance of the proposed algorithm and provide an analysis of its computational complexity. Finally, the conclusions of this article are summarized in Section IV.

II. PROBLEM STATEMENT AND ALGORITHMIC APPROACH

This section first introduces the TOA progression of ideal radar pulse sequence, then extends it to mixed radar pulse sequences. Based on this, an algorithm is proposed to deinterleave the noisy mixed radar pulse sequence with missing and spurious pulses.

A. Ideal Radar Pulse Sequence

Let t_k denote the arrival time of the k -th pulse, where $k = 1, 2, \dots, N$ and N represents the total number of pulses. The TOAs of the N pulses is represented as a set T

$$T = \{t_k\}_{k=1}^N, t_1 < t_2 < \dots < t_N \quad (1)$$

The ideal pulse sequences with different PRI modulations are shown in Fig. 1, and their TOA progression are introduced as follows.

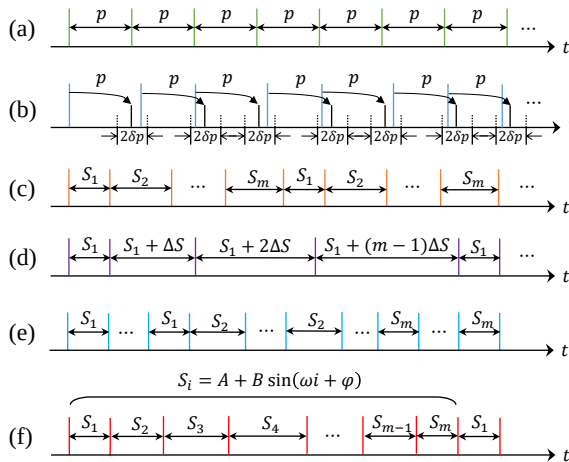


Fig. 1. Ideal radar pulse sequences: (a) fixed PRI; (b) jitter PRI; (c) staggered PRI; (d) sliding PRI; (e) D&S PRI; (f) wobulated PRI

Fixed PRI: The TOA differences of the fixed PRI pulse sequence are constant, i.e.,

$$t_{k+1} = t_k + p + n_k \quad (2)$$

where p represents the PRI value, and n_k represents the TOA measurement noise of the k -th pulse.

Jitter PRI: For the jitter PRI sequence, the average TOA difference remains constant, but a random jitter term is added, i.e.,

$$t_{k+1} = t_k + (1 + \delta_k)p + n_k \quad (3)$$

where $\delta_k \in [-\delta, \delta]$ generally following a uniform or normal distribution, and δ is the upper bound of the jitter term.

Staggered PRI: The staggered PRI sequence can be seen as multiple fixed PRI sequences with the same PRI value but different initial times. The identical PRI value is referred to as the frame period, while the intervals between different initial times are called the sub-PRI. Suppose a staggered PRI sequence consists of m sub-sequences, with the sub-PRI are represented by the set $S = \{S_i\}_{i=0}^{m-1}$, then the its TOA progression can be expressed as:

$$t_{k+1} = t_k + S_{\text{mod}(k,m)} + n_k \quad (4)$$

where $\text{mod}(\cdot, \cdot)$ denotes the modulo operation.

Sliding PRI: The sliding PRI sequence can be regarded as a type of staggered PRI sequence in which the sub-PRI vary uniformly. If a sliding PRI sequence has sub-PRI represented by the set $S = \{S_i\}_{i=0}^{m-1}$, then its TOA progression can be expressed as:

$$\begin{aligned} t_{k+1} &= t_k + S_{\text{mod}(k,m)} + n_k \\ S_{i+1} - S_i &= \text{const}, \forall 1 \leq i \leq m-1 \end{aligned} \quad (5)$$

D&S PRI: The D&S PRI sequence can also be regarded as a special type of staggered PRI sequence. In this sequence, the sub-PRI switches periodically among several fixed values, with each sub-PRI being repeated a certain number of times. If a D&S PRI sequence has sub-PRI represented by the set $S = \{S_i\}_{i=0}^{m-1}$, then its TOA progression can be expressed as:

$$\begin{aligned} S_0 &= S_1 = \dots = S_{l_1-1} \\ S_{l_1} &= \dots = S_{l_1+l_2-1} \\ &\vdots \\ S_{l_1+l_2+\dots+l_{K-1}} &= \dots = S_{l_1+l_2+\dots+l_K-1} \end{aligned} \quad (6)$$

$$t_{k+1} = t_k + S_{\text{mod}(k,m)} + n_k \quad (7)$$

where l_i represents the repetition count of each distinct sub-PRI, K is the number of distinct sub-PRI in the sequence and $l_1 + l_2 + \dots + l_K = m$.

Wobulated PRI: The wobulated PRI sequence can be regarded as a type of staggered PRI sequence in which the sub-PRI vary according to a sinusoidal function. If a wobulated PRI sequence has sub-PRI represented by the set $S = \{S_i\}_{i=0}^{m-1}$, then its TOA progression is expressed as:

$$\begin{aligned} t_{k+1} &= t_k + S_{\text{mod}(k,m)} + n_k \\ S_i &= A + B \sin(\omega i + \varphi) \end{aligned} \quad (8)$$

B. Mixed Radar Pulse Sequence

Compared to the ideal pulse sequence, the actual received mixed pulse sequence consists of multiple radar pulse sequences. The pulse sequences interleaved by multiple emitters is shown in Fig. 2. Let $T_i = \{T_k^i\}_{k=1}^{N_i}$ denotes the i -th pulse sequence. Due to the limitations of the reconnaissance receiver, it may fail to capture all pulses, leading to missing pulses in the i -th pulse sequence, which is denoted as T_{m_i} . Additionally, the reconnaissance receiver may also receive some spurious pulses that do not belong to any pulse sequence, denoted as $\tilde{T}$. Therefore, the TOAs of mixed radar pulse sequence can be represented as:

$$T = \left[\bigcup_{i=1}^K (T_i - T_{m_i}) \right] \cup \tilde{T} \quad (9)$$

where K represents the number of emitters in the mixed pulse sequence. Let $\xi = \frac{|T_{m_i}|}{|T_i|}$ and $\eta = \frac{|\tilde{T}|}{|T - \tilde{T}|}$, which denote the missing rate for each pulse sequence and the ratio of the number of spurious pulses to the number of actual pulses in the intercepted pulses, respectively. The notation $|\cdot|$ represents the cardinality of a set, i.e., the number of elements in the set.



Fig. 2. Mixed radar pulse sequences

C. Algorithm Workflow

The proposed deinterleaving algorithm for mixed radar pulse sequences consists of three core components. The first component is the PRI estimation module, which estimates the non-jitter PRI values by analyzing the PRI amplitude spectrum. When it comes to jitter PRI, the estimation is performed through segmenting the data. The second component is the pulse sequence retrieval module, which employs an improved DAG algorithm to separate the pulse sequences corresponding to the estimated PRI. For pulse sequences with complex PRI modulation, the improved DAG algorithm can retrieve all subsequences in a single process. The last component is the sub-PRIs estimation module, in which the phase information of the PRI spectrum is utilized. The operational process of the proposed algorithm is outlined below.

1) Calculate the PRI amplitude spectrum based on the TOA sequence $T = \{t_k\}_{k=1}^N$. Compared it to the detection threshold, a set of potential non-jitter PRIs P_1 can be identified. If P_1 is empty, proceed to step (2); otherwise, determine the estimated PRI for this iteration according to the PRI determination criterion and proceed to step (3).

2) Divide the TOA sequence into segments. For each segment, the PRI amplitude spectrum is calculated, and the results are summed across all segments to obtain the

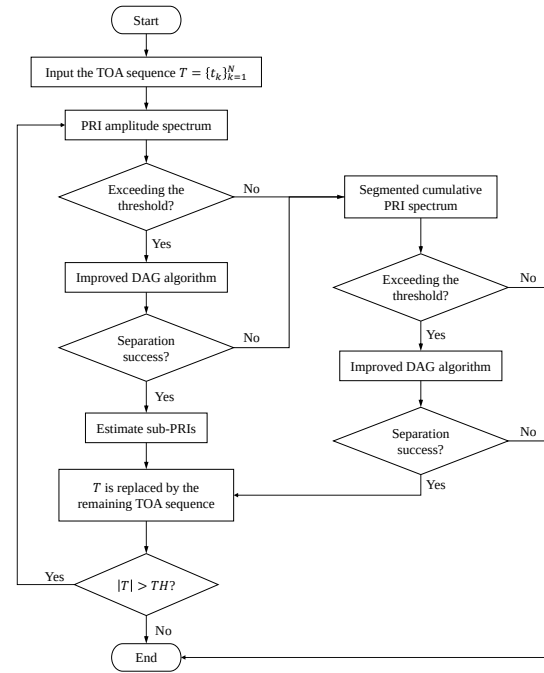


Fig. 3. Workflow of the proposed algorithm.

segmented cumulative PRI spectrum. Compared it with the detection threshold, a set of potential jitter PRIs P_2 can be identified. If P_2 is empty, the algorithm terminates; otherwise, proceed to step (5).

3) Separate the sequences corresponding to the PRI value estimated in step (1) through the improved DAG algorithm. If the separation meets the success criteria, the original set T is replaced by the remaining TOA sequence, and the algorithm continues to step (4). Otherwise, the algorithm reverts to step (2).

4) Estimate the sub-PRIs of complex PRI modulation based on the estimated PRI and sequence retrieval results. If the number of remaining pulses exceeds the deinterleaving threshold TH , the algorithm returns to step (1). Otherwise, the algorithm terminates.

5) Separate the sequences corresponding to the PRI value estimated in step (2) with the improved DAG algorithm. If the separation fails, the algorithm terminates. Otherwise, evaluate the number of remaining pulses: if it exceeds the deinterleaving threshold TH , proceed to step (1) for a new iteration of deinterleaving; otherwise, the algorithm terminates.

A detailed depiction of the proposed algorithm's workflow is provided in Fig. 3.

D. PRI Estimation Algorithm

In complex electromagnetic environments, accurately estimating the potential PRI values in mixed radar pulse sequences is a challenging task. To address this, this

section introduces a novel PRI estimation algorithm, which consists of two core components: the calculation of the PRI spectrum and the determination of the detection threshold. First, the concept of the PRI spectrum is defined, followed by an analysis of its characteristics for typical radar pulse sequences to demonstrate the algorithm's effectiveness. Next, the method for calculating the detection threshold is described in detail. Finally, the complete workflow of the proposed PRI estimation algorithm is summarized.

1. PRI Spectrum

Let $[\tau_{\min}, \tau_{\max}]$ be the range of the PRI to be investigated. This range is sampled at small intervals to generate a set of PRI candidate values Q . For a given TOA sequence $T = \{t_k\}_{k=1}^N$, its PRI spectrum is defined as follows:

$$I(q) = \sum_{k=1}^N \exp \left[j2\pi \frac{t_k}{q} \right] \quad (10)$$

where $q \in Q$ represents a possible PRI value. We refer to $|I(q)|$ as the PRI amplitude spectrum and $\text{Arg}[I(q)]$ as the PRI phase spectrum. Here, $|\cdot|$ denotes the modulus of a complex number and $\text{Arg}[\cdot]$ represents the principal value of the argument of a complex number. In this paper, the principal value is defined within the range $[0, 2\pi)$.

For a mixed pulse sequence $T = \bigcup_{i=1}^K T_i$, represented as the union of multiple subsets T_i , The PRI spectrum of T can be expressed as the sum of the PRI spectrum of T_i , i.e.,

$$I(q) = \sum_{i=1}^K I_i(q) \quad (11)$$

where $I(q)$ and $I_i(q)$ represent the PRI spectrum of T and T_i .

2. The PRI Spectrum of Typical Pulse Sequences

Since the PRI spectrum exhibits additivity for subsets, understanding its characteristics in typical pulse sequences is crucial. This subsection first examines the PRI spectral properties of an ideal fixed PRI sequence and then investigates the effects of TOA measurement noise, missing pulses, spurious pulses and PRI jitter. Building on these analyses, the PRI spectrum of complex PRI-modulated sequences is further explored.

Ideal sequence with fixed PRI. For an ideal sequence with fixed PRI p , the TOA of the k -th pulse is given by $t_k = t_1 + (k-1)p$. Consequently, its PRI spectrum can be expressed as:

$$\begin{aligned} I(q) &= \exp \left[j2\pi \frac{t_1}{q} \right] \sum_{k=1}^N \exp \left[j2\pi \frac{(k-1)p}{q} \right] \\ &= \exp \left[j2\pi \frac{t_1}{q} \right] \frac{1 - \exp \left[j2\pi N \frac{p}{q} \right]}{1 - \exp \left[j2\pi \frac{p}{q} \right]} \end{aligned} \quad (12)$$

$$|I(q)| = \left| \frac{\sin \left(N\pi \frac{p}{q} \right)}{\sin \left(\pi \frac{p}{q} \right)} \right| \quad (13)$$

Therefore, $|I(q)|$ reaches its maximum when $\frac{p}{q} \in \mathbb{Z}$, i.e., q is a divisor of p , which can then be used to determine p .

Meanwhile, note that

$$\lim_{x \rightarrow \mathbb{Z}} \frac{1 - \exp[j2\pi Nx]}{1 - \exp[j2\pi x]} = N \quad (14)$$

Thus, the information about the first TOA of the sequence, t_1 , is encoded in $\text{Arg}[I(p)]$.

Impact of TOA measurement noise. Assume that the TOA measurement noise for each pulse n_k is independent and identically distributed, following a uniform distribution within the range $[-a, a]$, i.e., $n_k \stackrel{\text{i.i.d.}}{\sim} \mathcal{U}(-a, a)$. Under this assumption, and the spectrum of the fixed PRI sequence is modified as follows:

$$I(q) = \exp \left[j2\pi \frac{t_1}{q} \right] \sum_{k=1}^N \exp \left[j2\pi \frac{(k-1)p + n_k}{q} \right] \quad (15)$$

The expected value and variance of (15) is given by

$$\begin{aligned} \mathbb{E}\{I(q)\} &= \exp \left[j2\pi \frac{t_1}{q} \right] \frac{q}{2\pi a} \sin \left(\frac{2\pi a}{q} \right) \\ &\quad \times \frac{1 - \exp \left[j2\pi N \frac{p}{q} \right]}{1 - \exp \left[j2\pi \frac{p}{q} \right]} \end{aligned} \quad (16)$$

$$\begin{aligned} \text{Var}\{I(q)\} &= \mathbb{E}\{I(q)\overline{I(q)}\} - |\mathbb{E}\{I(q)\}|^2 \\ &= N \left[1 + \frac{q^2}{8\pi^2 a^2} \cos \left(\frac{4\pi a}{q} \right) - \frac{q^2}{8\pi^2 a^2} \right] \end{aligned} \quad (17)$$

For small TOA measurement noise, i.e., $a \ll q$, we have $\text{Var}\{I(q)\} \rightarrow 0$, implying that $I(q)$ converges in probability to $\mathbb{E}\{I(q)\}$. Additionally, under this condition, $\frac{q}{2\pi a} \sin \left(\frac{2\pi a}{q} \right) \rightarrow 1$, demonstrating that small measurement noise has negligible impact on the peak position of the PRI spectrum, which remains at $\frac{p}{q} \in \mathbb{Z}$. The information of first TOA remains encoded in the phase of the PRI spectrum. This indicates that the algorithm maintains its performance even with measurement noise, showcasing its robustness.

Impact of pulse missing. Assume that each pulse is independently missed with probability ξ . Let X_k indicate whether the k -th pulse is present ($X_k = 1$) or missing ($X_k = 0$). Therefore, $X_1, X_2, \dots, X_N \stackrel{\text{i.i.d.}}{\sim} \mathcal{B}(1, 1 - \xi)$. Under this assumption, the spectrum of the ideal fixed PRI sequence is modified as follows:

$$I(q) = \exp \left[j2\pi \frac{t_1}{q} \right] \sum_{k=1}^N X_k \exp \left[j2\pi \frac{(k-1)p}{q} \right] \quad (18)$$

The expected value and variance of (18) is given by

$$E\{I(q)\} = \exp\left[j2\pi\frac{t_1}{q}\right] (1 - \xi) \frac{1 - \exp\left[j2\pi N\frac{p}{q}\right]}{1 - \exp\left[j2\pi\frac{p}{q}\right]} \quad (19)$$

$$\text{Var}\{I(q)\} = N\xi - N\xi^2 \quad (20)$$

At the peak points of the original PRI spectrum, $\{q \mid p/q \in \mathbb{Z}\}$, it follows that:

$$I(q) = \exp\left\{j2\pi\frac{t_1}{q}\right\} \sum_{k=1}^N X_k \quad (21)$$

Thus, the impact of missing pulses on the peak points of the PRI spectrum is reduced solely to their amplitude, which equals the number of pulses actually intercepted, while leaving their phase unaffected. This demonstrates the robustness of the algorithm to pulse missing.

Impact of spurious pulses. Let $M(t)$ represent the number of spurious pulses in the time interval $(0, t]$. The random process $M(t), t > 0$ can be modeled as a Poisson process [7]. The event times in a Poisson process, when conditioned on the total number of events, follow the same distribution as the order statistics of a uniform distribution [25].

Thus, if M spurious pulses are received during the observation period T , with their TOAs given by $\{t_k\}_{k=1}^M$, it follows that $t_1, t_2, \dots, t_M \stackrel{\text{i.i.d.}}{\sim} \mathcal{U}(0, T)$. By defining $X_k = t_k - \frac{T}{2}$, it can be shown that $X_1, X_2, \dots, X_M \stackrel{\text{i.i.d.}}{\sim} \mathcal{U}(-\frac{T}{2}, \frac{T}{2})$. Consequently, the PRI spectrum of the spurious pulses can be expressed as follows:

$$\begin{aligned} I &= \sum_{k=1}^M \exp\left[j2\pi\frac{T_k}{q}\right] \\ &= \exp\left[j\pi\frac{T}{q}\right] \sum_{k=1}^M \exp\left[j2\pi\frac{X_k}{q}\right] \end{aligned} \quad (22)$$

The expected value and variance of (22) is given by

$$E\{I(q)\} = \exp\left[j2\pi\frac{T}{q}\right] \frac{Mq}{\pi T} \sin\left(\frac{\pi T}{q}\right) \quad (23)$$

$$\text{Var}\{I(q)\} = M \left[-\frac{q^2}{2\pi^2 T^2} + \frac{q^2}{2\pi^2 T^2} \cos\left(\frac{2\pi T}{q}\right) + 1 \right] \quad (24)$$

In general, the PRI candidate q is much smaller than the observation period T , i.e., $q \ll T$. The variance $\text{Var}\{I(q)\}$ approaches M and significantly exceeds the mean $|E\{I(q)\}|$. This dominance of variance indicates the PRI spectrum of spurious pulses on the PRI spectrum is highly stochastic.

However, when a fixed PRI sequence is mixed with spurious pulses, based on the additivity of the mathematical expectation and variance, the expected value of its PRI spectrum at the peak point is N , and the variance is M , where N is the number of pulses in the fixed PRI sequence. To assess the impact of spurious pulses on the PRI spectrum peak, the coefficient of variation (CV),

which is defined as the ratio of the standard deviation to the mean, is computed:

$$\text{CV} = \frac{\sqrt{M}}{N} \quad (25)$$

In general, if the CV of a random variable is less than 0.1, it can be considered to have a small degree of dispersion. Therefore, when $M/N \leq 0.01N$, the impact of false pulses on the PRI spectrum peak can be considered negligible. For example, if $N = 100$, then when $M/N \leq 1$, i.e., when the number of spurious pulses is smaller than the number of true pulses, their effect on the PRI spectrum peak is negligible. Consequently, the algorithm is robust against spurious pulses.

Impact of PRI jitter. Assuming that the jitter amounts in a jitter PRI sequence are independent and identically distributed, following a Gaussian distribution with zero mean, i.e., $\delta_1, \delta_2, \dots, \delta_N \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$, the spectrum of the jitter PRI sequence is given by:

$$\begin{aligned} I(q) &= \exp[j2\pi t_1] \\ &\times \sum_{k=1}^N \exp\left\{j2\pi\frac{1}{q} \left[(k-1)p + p \sum_{m=1}^{k-1} \delta_m \right] \right\} \end{aligned} \quad (26)$$

In this case, the phase of the PRI spectrum provides limited useful information. As a result, we focus on computing the expected PRI amplitude spectrum.

$$\begin{aligned} E\{|I(q)|\} &= N + 2 \sum_{k=1}^{N-1} (N-k) \exp\left[-k\frac{2\pi^2 p^2}{q^2} \sigma^2\right] \\ &\times \cos\left(k2\pi\frac{p}{q}\right) \end{aligned} \quad (27)$$

The result from the above summation is complex. To simplify it, we introduce the following notations:

Let $x = \frac{p}{q}$, and define

$$f(m, n, x) = \exp(-m2\pi^2 \sigma^2 x) \cos(n2\pi x) \quad (28)$$

$$\begin{aligned} g_1 &= 2f(2, 0, x) [2 - f(1, 1, x) - f(-1, 1, x) \\ &\quad + f(N+1, N-1, x) + f(N-1, N+1, x) \\ &\quad - 2f(N, N, x)] \end{aligned} \quad (29)$$

$$g_2 = f(2, 0, x) - 2f(1, 1, x) + 1 \quad (30)$$

$$g_3 = 2N [f(1, 1, x) - f(2, 0, x)] \quad (31)$$

Then, (27) can be expressed as:

$$E\{|I(q)|\} = \frac{g_1}{g_2} + \frac{g_3}{g_2} + N \quad (32)$$

The above expression continues to exhibit peaks near $\{q \mid \frac{p}{q} \in \mathbb{Z}\}$. Fig. 4 shows the expected PRI amplitude spectrum of jitter PRI with different σ values when N equals 100.

However, these peaks are accompanied by significant variance, making them less distinct. To address this issue,

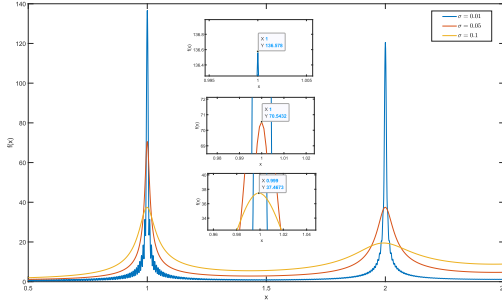


Fig. 4. The expected PRI amplitude spectrum of jitter PRI with different σ values.

a method is proposed to reduce the variance by segmenting the TOA sequence and summing the amplitude spectra of each segment. Specifically:

$$\tilde{I}(q) = \sum_{i=1}^{\lfloor T/L \rfloor} \left| \sum_{(i-1)L \leq t_k < iL} \exp \left[j2\pi \frac{t_k}{q} \right] \right| \quad (33)$$

where L denotes the segment length, and $\lfloor \cdot \rfloor$ represents the floor operation. As the summation term in (27) diminishes to 0 with increasing k , the contribution of later terms becomes negligible. This ensures that the segmentation approach effectively reduces variance while preserving the primary characteristics of the PRI spectrum.

Sequence with complex PRI modulation. Building on the previous analysis of PRI sequences, a complex PRI-modulated sequence can be represented as a combination of multiple fixed PRI subsequences, each with a unique starting time. Suppose that the sequence consists of m subsequences and has a frame period p . Thus, its PRI spectrum can be expressed as:

$$\begin{aligned} I(q) &= \sum_{i=1}^m I_i(q) \\ &= \sum_{i=1}^m \exp \left[j2\pi \frac{t_1^{(i)}}{q} \right] \frac{1 - \exp \left[j2\pi N_i \frac{p}{q} \right]}{1 - \exp \left[j2\pi \frac{p}{q} \right]} \end{aligned} \quad (34)$$

where N_i and $t_1^{(i)}$ represent the number of pulses and the first TOA of the i -th subsequence. Therefore, the algorithm has limitations when applied to complex PRI modulations. Specifically, as the differences between sub-PRIs decrease, the likelihood of correctly identifying the frame period diminishes. In the extreme scenario where all sub-PRIs are identical, the PRI amplitude spectrum's response at the frame period nearly vanishes. Under such circumstances, the algorithm becomes nearly ineffective in detecting the frame period. However, this is a common issue for TOA-based algorithms.

3. Threshold for Detection of PRIs

To detect the PRI from the PRI amplitude spectrum, it is essential to differentiate the true PRI from other

candidates. This differentiation relies on three key criteria: the observation time criterion, the harmonic elimination criterion, and the noise elimination criterion. The thresholds for detection of PRIs can be established based on these criteria.

Observation time criterion. If a fixed PRI sequence with a PRI of p exists throughout the entire observation time T , the theoretical number of pulses would be T/p . However, in practical situations, the fixed PRI sequence may not persist throughout the entire observation period, and pulse missing may occur. Thus, we define the following criterion with a margin of tolerance:

$$|I(q)| \geq \alpha \frac{T}{q} \quad (35)$$

where α is a tunable parameter that needs to be adjusted based on the pulse miss rate ξ , with the condition that $\alpha \leq 1 - \xi$ to ensure proper performance. If α is set too high, the detector becomes overly strict and may fail to detect valid PRI sequences containing missing pulses. Conversely, if α is too low, the detector becomes overly permissive, increasing the likelihood of spurious sequences being accepted. Therefore, α should be carefully chosen to balance robustness to missing pulses and resistance to noise. In practice, α is typically set to 0.5.

Harmonic elimination criterion. The PRI amplitude spectrum exhibits peaks at $q \mid p/q \in \mathbb{Z}$. These harmonics can interfere with PRI detection. To eliminate their influence, whether a sequence with the corresponding PRI exists in the mixed sequence is determined using the following formula:

$$|I(q)| > m|I(mq)|, m \in \mathbb{Z} \quad (36)$$

Noise elimination criterion. To mitigate the impact of spurious pulses on PRI detection results, a constant false alarm rate (CFAR) detection algorithm is adopted to determine whether the mixed pulse sequence contains a sequence with the corresponding PRI. In this study, the commonly used CA-CFAR detector is applied due to its simplicity and effectiveness in handling interference. Assuming the detection threshold of the CFAR detector at q is $C(q)$, we can judge whether q is a PRI by the criterion:

$$|I(q)| > C(q) \quad (37)$$

Based on the aforementioned three criteria, the threshold is determined as follows:

$$H(q) = \max \left\{ \alpha \frac{T}{q}, m|I(mq)|, C(q) \right\} \quad (38)$$

From the amplitude spectrum, the PRI with the largest ratio of peak value to threshold is selected as the estimated value $\hat{p}$ for this iteration:

$$\hat{p} = \arg \max_{q \in P} \frac{|I(q)|}{H(q)} \quad (39)$$

where P is the set of PRI values corresponding to the peaks in the PRI amplitude spectrum that exceed the detection threshold.

4. Workflow of the PRI Estimation Algorithm

The workflow of the PRI estimation algorithm is outlined below:

Input:

- $\{t_k\}_{k=1}^N$: TOA sequence
- Q : set of possible PRI values
- L : segment length

Output:

- Estimated PRI value $\hat{p}$ for this iteration

Step 1: Compute the PRI amplitude spectrum $|I(q)|$ using (33).

Step 2: Calculate the detection threshold $H(q)$ using (38).

Step 3: Select the estimated PRI value $\hat{p}$ for this iteration using (39).

E. Sequence Retrieval Algorithm

After identifying the potential PRI values, the next step is to separate the corresponding sequence. Xie et al. [19] proposed a graph theory-based method that models pulses as vertices and pulse pairs as edges, transforming the sequence separation problem to extracting the longest path from a DAG. By leveraging global information, this approach achieves high accuracy and effectiveness in sequence separation.

However, the method proposed by Xie et al. is limited to sequences with fixed PRI and is not generalizable to different PRI types. In this subsection, an improved method that extends the algorithm to enable the separation of sequences with complex PRI modulations is proposed.

In a DAG, the indegree of a node represents the number of edges pointing to that node, while the outdegree represents the number of edges originating from that node. By utilizing the indegree and outdegree information of nodes, the “longest path extraction” in [19] is improved to “top-k longest paths extraction” enabling effective separation of sequences with complex PRI modulation. The workflow of the improved algorithm is as follows:

Input:

- $\{t_k\}_{k=1}^N$: TOA sequence
- $\hat{p}$: estimated PRI value
- H_1 : threshold for retrieval success
- H_2 : threshold for path segmentation

Output:

- TOA sequence $T_{\hat{p}}$ corresponding to $\hat{p}$

Step 1: Using the TOA sequence $\{t_k\}_{k=1}^N$ and the estimated PRI $\hat{p}$, construct the DAG as described in Algorithm 1. The resulting DAG is an unweighted graph, and its nodes can be topologically sorted as $(1, 2, \dots, N)$.

Step 2: For each node in the DAG, calculate its indegree and outdegree. Mark nodes with zero indegree or zero outdegree for further processing.

Step 3: For each node with zero outdegree, i.e., terminal nodes in the graph, use Algorithm 2 to compute the longest path ending at this node, obtaining a path set $\{P_k\}_{k=1}^M$.

Step 4: Group the paths in $\{P_k\}_{k=1}^M$ by their starting node, where each starting node has zero indegree. In each group, retain only the longest path, producing a reduced set of paths denoted as $\{P'_k\}_{k=1}^{M'}$.

Step 5: Check if the retrieval is successful by comparing the length of the longest path in $\{P'_k\}_{k=1}^{M'}$ with the threshold H_1 . If the length exceeds H_1 , proceed to the next step. Otherwise, consider the retrieval unsuccessful, set $T_{\hat{p}} = \emptyset$, and terminate the algorithm.

Step 6: Sort the paths in $\{P'_k\}_{k=1}^{M'}$ in ascending order of path length. Compute the difference in length between consecutive paths, defined as the length of the latter path minus that of the former. Identify the largest difference. If the largest difference exceeds the threshold H_2 , remove all paths that precede the path corresponding to this largest difference. Otherwise, retain all paths.

Finally, the nodes in $\{P'_k\}_{k=1}^{M'}$ correspond to the results of the sequence retrieval. For complex modulated PRI sequences, each path represents a subsequence.

Algorithm 1: Construction of DAG

Input: $\{t_k\}_{k=1}^N$: TOA sequence, $\hat{p}$: Estimated PRI, δ : Jitter bound, K : Number of pulses to extend;

Output: $W = (w_{ij})_{N \times N}$: Adjacency matrix;

```

1  $A = \bigcup_{k=1}^K [k(1 - \delta)\hat{p}, k(1 + \delta)\hat{p}]$ ;
2 FOR  $i = 1 : N$  DO
3   FOR  $j = i + 1 : N$  DO
4     IF  $t_j - t_i \in A$  THEN
5        $W(i, j) = 1$ ;
6     ELSE
7        $W(i, j) = 0$ ;
8     END
9   END
10 END
```

The quality and reliability of the retrieved sequences depend not only on the structure of the DAG and the path extraction process, but also critically on the choice of retrieval thresholds.

In particular, the threshold H_1 plays a key role in determining whether candidate sequences satisfy the retrieval criterion. If H_1 is set too high, valid sequences may be rejected, leading to missed detections. Conversely, a low H_1 increases the risk of spurious sequences being accepted, resulting in false alarms. In practice, H_1 is typically set to the right-hand side of Equation (35), i.e., $\alpha T / \hat{p}$.

The threshold H_2 serves to suppress spurious paths during the retrieval of subsequences in complex PRI-

Algorithm 2: Longest Path in DAG with Fixed Endpoint

Input: $W = (w_{ij})_{N \times N}$: Adjacency matrix, v : Endpoint
Output: **Len**: The length of the longest path;
Path: The longest path;

```

1  $L = \text{zeros}(N, 1)$ ;
2  $L(v) = 1$ ;
3 Successor =  $\text{zeros}(N, 1)$ ;
4 FOR  $i = v - 1 : -1 : 1$  DO
5   FOR  $j = i + 1 : v$  DO
6     IF  $W(i, j) == 1$  &  $L(j) \geq L(i)$ 
7       THEN
8          $L(i) = L(j) + 1$ ;
9         Successor( $i$ ) =  $j$ ;
10      END
11   END
12 Len =  $\max(L)$ ;
13 Start =  $\text{find}(L == \text{Len})$ ;
14 Path = Start;
15 WHILE Successor(Start) > 0 DO
16   Start = Successor(Start);
17   Path = [Path, Start];
18 END

```

modulated signals. As genuine subsequences usually have substantially longer paths than spurious ones, the algorithm is relatively insensitive to the precise value of H_2 . Consequently, H_2 can be chosen loosely and generally has limited impact on performance. A typical setting is $H_2 = 0.5H_1$.

F. Estimation of Sub-PRIs

Based on the previous analysis, we can draw the following conclusions: First, the sequence with complex PRI modulation can be decomposed into multiple subsequences with different initial times. Second, the improved DAG algorithm can effectively identify and retrieve all these subsequences. Finally, the information about the initial time of the pulse sequence is encoded in the phase of the PRI spectrum. Therefore, we can calculate the PRI spectrum phases of the subsequences retrieved by the improved DAG algorithm to determine the initial times of each subsequence. And then, by performing differential analysis on the sorted initial times, we can accurately obtain the sub-PRIs in complex PRI modulation. The algorithm 3 presents the process for estimating the sub-PRIs.

III. SIMULATION EXPERIMENTS AND PERFORMANCE ANALYSIS

This section validates the theoretical analysis of the PRI spectrum and demonstrates the new algorithm's ca-

Algorithm 3: Sub-PRIs Estimation

Input: $\{t_k^{(1)}\}_{k=1}^{N_1} \dots \{t_k^{(m)}\}_{k=1}^{N_m}$: Sequence retrieved using DAG, $\hat{p}$: Estimated PRI;
Output: $S = \{S_k\}_{k=1}^m$: Estimated Sub-PRIs;

```

1 FOR  $i = 1 : m$  DO
2    $I = \sum_{k=1}^{N_i} \exp \{j2\pi t_k^{(i)} / \hat{p}\}$ ;
3    $P(i) = \hat{p} * \text{Arg}(I) / 2\pi$ ;
4 END
5  $P_s = \text{sort}(P, 'ascend')$ ;
6  $P_s(m+1) = P_s(1) + \hat{p}$ ;
7  $S = \text{diff}(P_s)$ ;

```

pability in deinterleaving mixed radar pulse sequences through simulation experiments. In Section III-A, we introduce the deinterleaving process of mixed radar pulse sequences with a specific example, followed by the simulation results at each step. Section III-B evaluates and analyzes the performance of the new algorithm under various conditions, including different missing rates, spurious pulse counts and PRI jitter bounds. Section III-C provides comparative experiments against baseline methods. Finally, Section III-D analyzes the algorithm's computational complexity.

A. Example of Deinterleaving Mixed Pulse Sequences

In the example, an interleaved pulse stream is composed of four target pulse sequences and spurious pulses, the specifications are listed in Table I. The steps involved in deinterleaving using the proposed algorithm are outlined below.

In the first deinterleaving iteration, the PRI amplitude spectrum of the mixed pulse sequence is calculated, as shown in Fig. 5(a). Based on the detection threshold and PRI determination criteria, the estimated PRI value for this iteration is 524.6 μs . The corresponding TOA sequence is then separated from the mixed sequence using the improved DAG algorithm. Fig. 5(b) displays the pulse separation results by plotting the points $(k, 2\pi t_k / \hat{p})_{k=1}^N$ on polar coordinates, where each group of differently colored points represents a path returned by the improved DAG algorithm and corresponds to a subsequence of the complex PRI modulation sequence. The phase at the estimated PRI for each subsequence is then calculated, and the sub-PRIs are extracted. The results are [35.090, 39.272, 42.612, 44.776, 44.442, 43.382, 39.740, 36.075, 31.097, 27.702, 25.264, 25.139, 26.658, 29.784, 33.566] μs .

The second and third deinterleaving iterations follow a process similar to the first iteration, and the corresponding results are shown in Fig. 5(c)-5(f). In the second iteration, the estimated PRI is 285 μs , and the estimated sub-PRIs are [28.326, 54.231, 87.874, 114.569] μs . In the third iteration, the estimated PRI is 371 μs .

TABLE I
Specifications of Mixed Pulse Sequences 1

	Sequence 1	Sequence 2	Sequence 3	Sequence 4
PRI modulation	Wobulated	Staggered	Fixed	Jitter
Frame period(μs)	524.55	285	-	-
Sub-PRIs(μs)	$\{35 + 10\sin(0.14\pi m)\}_{m=0}^{14}$	28,54,88,115	-	-
PRI value(μs)	-	-	371	463
Missing rate	20%	20%	20%	20%
Jitter bound	-	-	-	$\pm 15\%$
Spurious Pulses Count	150			
Noise error(μs)	± 2			
Observation time(μs)	60000			

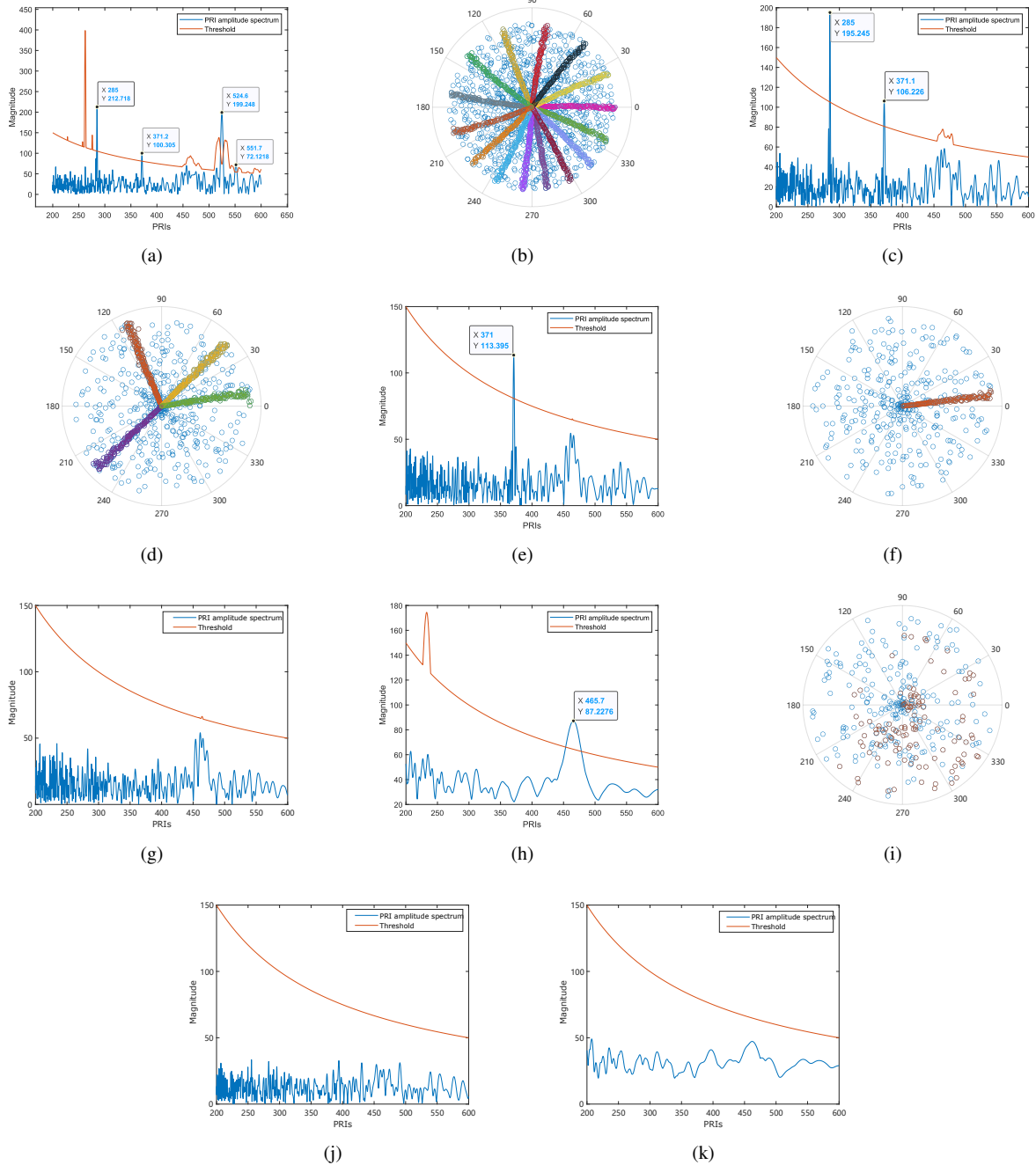


Fig. 5. Example of Deinterleaving Mixed Pulse Sequences

(a)-(b) First iteration of deinterleaving. (c)-(d) Second iteration of deinterleaving. (e)-(f) Third iteration of deinterleaving. (g)-(i) Fourth iteration of deinterleaving. (j)-(k) Fifth iteration of deinterleaving.

The deinterleaving results for the fourth and fifth iterations are shown in Fig. 5(g)-5(i). Since the remaining sequence has a jitter PRI, the segmented PRI amplitude spectrum works, and the estimated PRI is 465.7 μs . After the fifth iteration of deinterleaving, as shown in Fig. 5(j)-5(k), both the unsegmented and segmented PRI amplitude spectra could not extract the PRI value, leading to the termination of the algorithm.

The experimental results have demonstrated that the proposed algorithm can successfully deinterleave the mixed pulse sequences and estimate the sub-PRI of the sequence with complex PRI modulation under conditions involving TOA noise, pulse missing, and spurious pulses.

B. Performance Evaluation

To evaluate the performance of the proposed algorithm, a mixed sequence consisting of four TOA sequences and spurious pulses was selected for testing. The first pulse sequence has a wobulated, sliding or D&S PRI modulation. The second employs staggered PRI modulation. The third is a simple fixed PRI sequence. The fourth is a jitter PRI sequence. The specifications of the pulse sequences are provided in Table II.

The performance of the algorithm is evaluated based on two different metrics: merr, which measures the average error in PRI value estimation, and PRism, which assesses the accuracy of pulse retrieval. The definitions of the two metrics are as follows:

$$\text{merr} = \frac{1}{n} \sum_{i=1}^n \frac{|p_i - \hat{p}_i|}{p_i} \quad (40)$$

$$\text{PRism} = \frac{1}{n} \sum_{i=1}^n \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i} \times \frac{\text{TP}_i}{\text{TP}_i + \text{FN}_i} \quad (41)$$

where $\hat{p}_i$ represents the estimated PRI value, and p_i is the true PRI value. TP_i and FP_i represent the number of true pulses and spurious pulses in the i -th retrieved sequence, while FN_i represents the number of true pulses that are not retrieved.

Based on the simulation results presented in Fig. 6, we can draw the following conclusions:

1) With a jitter bound of 10%, merr remains below 0.25% as long as either the miss rate does not exceed 20% or the number of spurious pulses does not exceed 100. Since the median PRI value is 400 μs , this means that the average PRI estimation error does not exceed 1 μs , indicating that the algorithm achieves high accuracy in this scenario. As the electromagnetic environment becomes more complex, the algorithm continues to successfully deinterleave mixed pulse sequences, though with slightly reduced accuracy.

2) As the jitter bound, spurious pulse count, and missing rate increase, the algorithm's performance gradually deteriorates. PRI jitter disrupts the periodicity of the pulse sequence, leading to mutual interference between different pulse sequences. Spurious pulses and pulse

missing mainly affect the algorithm's performance by causing false alarms or missed alarms in PRI estimation. When both spurious pulses and pulse missing increase simultaneously, the missing pulses may coincide with the positions of the spurious pulses, leading to spurious pulses being mistakenly identified as legitimate pulses. This further impacts the algorithm's accuracy.

3) The impact of spurious pulses on the algorithm's performance is greater than that of pulse missing. This is mainly because pulse missing affects each pulse sequence independently, meaning that missing pulses in one sequence typically do not interfere with the deinterleaving of other sequences. In contrast, spurious pulses are always present in the mixed sequence to be deinterleaved, affecting the entire deinterleaving process.

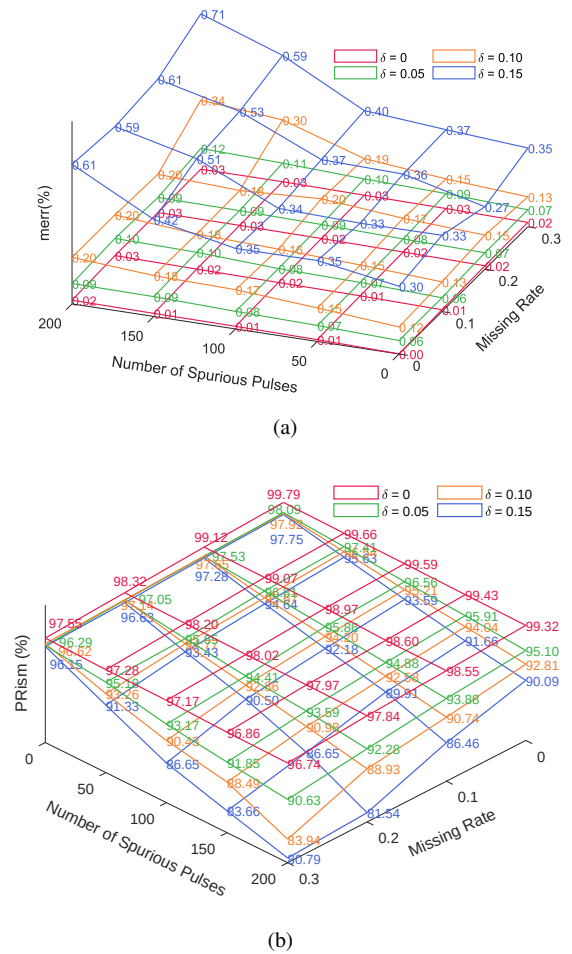


Fig. 6. Performance of the proposed algorithm. (a) merr. (b) PRism.

C. Comparative Simulation Experiments

In 2021, Chen et al. [20] proposed a mixed radar pulse sequence deinterleaving algorithm that integrates correlation matching, improved PRI transform, and clustering-based sequence retrieval method. This algorithm, named

TABLE II
Specifications of Mixed Pulse Sequences 2

	Sequence 1	Sequence 2	Sequence 3	Sequence 4
PRI modulation	Wobulated/Sliding/D&S	Staggered	Fixed	Jitter
Frame period(μs)	400-600	200-600	-	-
Number of Sub-PRIs	7-15	3-5	-	-
PRI value(μs)	-	-	200-600	200-600
Missing rate	0-30%	0-30%	0-30%	0-30%
Jitter bound	-	-	-	0- $\pm 15\%$
Spurious Pulses Count	0-200			
Noise error(μs)	± 2			
Observation time(μs)	60000			

TABLE III
Specifications of Mixed Pulse Sequences 3

	Sequence 1	Sequence 2	Sequence 3
PRI modulation	Staggered	Fixed	Jitter
Frame period(μs)	30-100	-	-
Number of Sub-PRIs	2-3	-	-
PRI value(μs)	-	20-100	20-100
Missing rate	0-30%	0-30%	0-30%
Jitter bound	-	-	$\pm 10\%$
Spurious Pulses Count	0-150		
Noise error(μs)	± 2		
Observation time(μs)	10000		

Clustering-Matching Transform (CMT), serves as the baseline for performance comparison.

The specifications of the mixed pulse sequences are presented in Table III. The deinterleaving performance of the mixed pulse sequence under different spurious pulse counts and pulse missing rates for both algorithms is displayed in Fig. 7. As shown in Fig. 7, when the spurious pulse count and missing rate are low, the performance of both algorithms is comparable. However, as the spurious pulse count and missing rate increase, the performance of the proposed algorithm outperforms that of CMT, which validates its superiority in deinterleaving mixed radar pulse sequences.

To further evaluate the performance of the proposed algorithm, we conduct comparative experiments using mixed sequences comprising multiple jitter-PRI signals. The improved SDIF (ISDIF) algorithm proposed in [4] is adopted as a baseline for comparison. In our simulations, we generate mixed pulse sequences consisting of three jitter-PRI sequences, with their specifications summarized in Table IV. The simulation results are shown in Fig. 8. As shown in the figure, the proposed algorithm exhibits superior capability in deinterleaving such mixed jitter-PRI sequences.

D. Algorithm complexity

This subsection presents a comparison of the computational costs between the proposed method and CMT, as shown in Table V.

In the proposed method, assuming the number of pulses in the mixed pulse sequence is N and the number

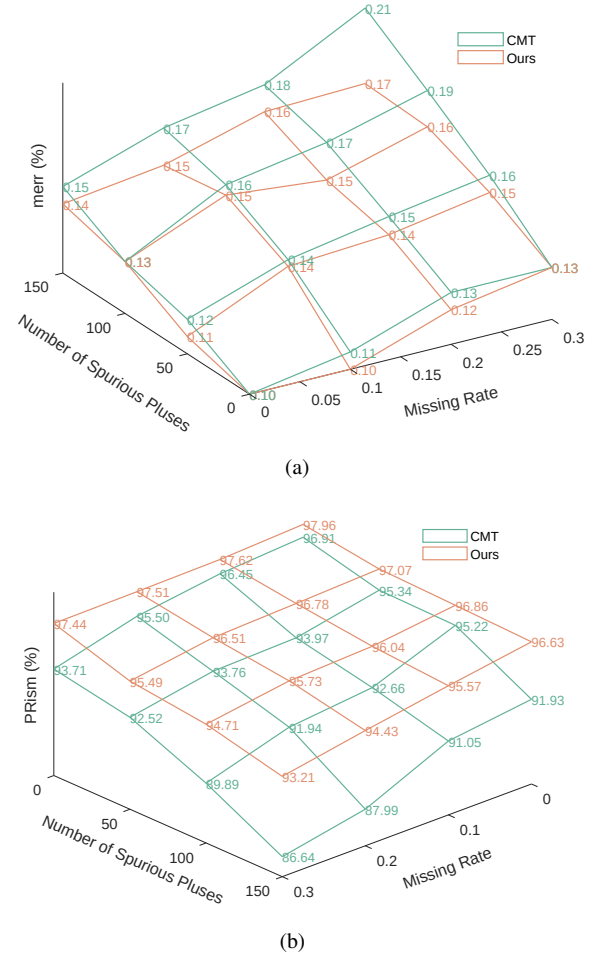


Fig. 7. Performances of two different algorithms.
(a) merr. (b) PRISM.

of PRI candidates is M , the computational complexity of the PRI estimation stage is as follows: For each PRI candidate, it is necessary to sum N complex numbers to obtain the PRI spectrum. Therefore, the computational complexity for this stage is $O(MN)$. In the pulse sequence retrieval stage, the adjacency matrix of the DAG needs to be calculated by checking the TOA difference between every two pulses. This operation has a complexity of $O(N^2)$. Additionally, in the algorithm for finding the longest path with fixed endpoints, the adjacency matrix must be traversed, which also has a complexity of $O(N^2)$.

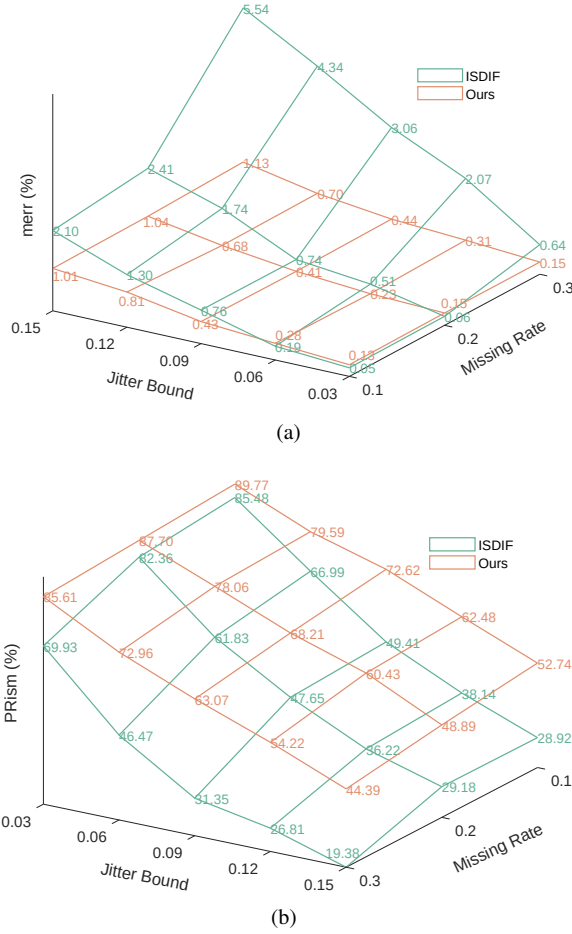


Fig. 8. Performances of two different algorithms. (a) merr. (b) Prism.

TABLE IV
Specifications of Mixed Pulse Sequences 4

	Specifications
Number of emitters	3
PRI modulation	Jitter
PRI value(μs)	200-600
Jitter bound	$\pm 3 - \pm 15\%$
Missing rate	10%-30%
Observation time(μs)	60000

Hence, the overall complexity of the pulse sequence retrieval process is $O(N^2)$. In the sub-PRI estimation step, the phase corresponding to each sub-sequence at the estimated PRI is calculated. If there are K sub-sequences, the computational complexity for this step is $O(KN)$. Since typically $M \gg K$ and $N \gg K$, the overall computational complexity for one iteration deinterleaving is $O(MN + N^2)$.

In CMT, the PRI estimation algorithm employs both CMM and the improved PRI transform, with computational complexities of $O(q_{max}LM)$ and $O(N^2)$, respectively, where L is the length of the zero-one sequence, and q_{max} is the largest PRI candidate. The complexity of the sequence retrieval algorithm in CMT is $O(N^2)$, and the computational complexity for sub-PRI estimation

TABLE V
Computational Costs of Two Deinterleaving Algorithms

	CMT	Ours
PRI estimation	$O(q_{max}LM)$	$O(MN)$
Sequence retrieval	$O(N^2)$	$O(N^2)$
Sub-PRI estimation	$O(K)$	$O(KN)$
One iteration of deinterleaving	$O(q_{max}LM)$	$O(MN + N^2)$

is $O(K)$. Since $q_{max}LM > N^2 \gg K$ typically holds, the overall computational complexity for one iteration of deinterleaving is $O(q_{max}LM)$.

As $q_{max}LM$ is generally greater than both MN and N^2 , the proposed algorithm demonstrates lower computational complexity compared to CMT for one iteration of deinterleaving. Additionally, for PRI sequences with complex modulation, the proposed algorithm can separate all sub-sequences in just one iteration, whereas CMT requires multiple iterations to achieve the same result. Therefore, the proposed algorithm can complete the deinterleaving of mixed pulse sequences with fewer iterations compared to CMT.

IV. CONCLUSION

This paper presents a novel algorithm for deinterleaving mixed pulse sequences under conditions involving TOA noise, spurious pulses, and missing pulses. The algorithm generates a spectrum in the PRI domain, referred to as the PRI spectrum. The amplitude information of the PRI spectrum is used to estimate the PRI, while the phase information is used to estimate the sub-PRI in complex PRI modulation. Additionally, an improved pulse sequence retrieval method is proposed to separate mixed pulse sequences. By leveraging the in-degree and out-degree information of nodes in DAG, the original DAG-based sequence retrieval algorithm is extended to accommodate different PRI types. Simulation results demonstrate that the proposed algorithm can effectively deinterleave mixed radar pulse sequences. When the jitter bound is 15%, the miss rate is 30%, and the spurious pulse count is 200 within 60 milliseconds, merr and PRism achieved 0.71% and 80.79%, respectively, demonstrating significant superiority.

Based on the analysis in Sections II-D, relying solely on a TOA parameter has some limitations when deinterleaving mixed radar sequences. Therefore, multi-parameter joint deinterleaving is an important research direction.

REFERENCES

- [1] H. Mardia
New techniques for the deinterleaving of repetitive sequences
In *IEE Proceedings F (Radar and Signal Processing)*, vol. 136, no. 4. IET, 1989, pp. 149–154.
- [2] C. Davies and P. Hollands
Automatic processing for esm

- In *IEEE Proceedings F (Communications, Radar and Signal Processing)*, vol. 129, no. 3, IET, 1982, pp. 164–171.
- [3] D. Milojević and B. M. Popović
Improved algorithm for the deinterleaving of radar pulses
In *IEEE Proceedings F (Radar and Signal Processing)*, vol. 139, no. 1, IET, 1992, pp. 98–104.
 - [4] Y. Liu and Q. Zhang
Improved method for deinterleaving radar signals and estimating pri values
IET Radar, Sonar & Navigation, vol. 12, no. 5, pp. 506–514, 2018.
 - [5] P. S. Ray
A novel pulse toa analysis technique for radar identification
IEEE Transactions on Aerospace and Electronic systems, vol. 34, no. 3, pp. 716–721, 1998.
 - [6] D. Nelson
Special purpose correlation functions for improved signal detection and parameter estimation
In *1993 IEEE International Conference on Acoustics, Speech, and Signal Processing*, vol. 4, IEEE, 1993, pp. 73–76.
 - [7] K. Nishiguchi and M. Kobayashi
Improved algorithm for estimating pulse repetition intervals
IEEE transactions on Aerospace and Electronic Systems, vol. 36, no. 2, pp. 407–421, 2000.
 - [8] R. J. Orsi, J. B. Moore, and R. E. Mahony
Interleaved pulse train spectrum estimation
In *Fourth International Symposium on Signal Processing and Its Applications*, vol. 1, IEEE, 1996, pp. 125–128.
 - [9] W. A. Sethares and T. W. Staley
Periodicity transforms
IEEE transactions on Signal Processing, vol. 47, no. 11, pp. 2953–2964, 1999.
 - [10] D. D. Muresan and T. W. Parks
Orthogonal, exactly periodic subspace decomposition
IEEE Transactions on Signal Processing, vol. 51, no. 9, pp. 2270–2279, 2003.
 - [11] S.-C. Pei and K.-W. Chang
Closed-form orthogonal ramanujan integer basis
IEEE Signal Processing Letters, vol. 24, no. 1, pp. 116–120, 2016.
 - [12] J.-W. Tao, C.-Z. Yang, and C.-W. Xu
Estimation of pri stagger in case of missing observations
IEEE Transactions on Geoscience and Remote Sensing, vol. 58, no. 11, pp. 7982–8001, 2020.
 - [13] J. B. Moore and V. Krishnamurthy
Deinterleaving pulse trains using discrete-time stochastic dynamic-linear models
IEEE transactions on signal processing, vol. 42, no. 11, pp. 3092–3103, 1994.
 - [14] T. Conroy and J. B. Moore
The limits of extended kalman filtering for pulse train deinterleaving
IEEE transactions on signal processing, vol. 46, no. 12, pp. 3326–3332, 1998.
 - [15] T. L. Conroy and J. B. Moore
On the estimation of interleaved pulse train phases
IEEE transactions on signal processing, vol. 48, no. 12, pp. 3420–3425, 2000.
 - [16] A. Logothetis and V. Krishnamurthy
An interval-amplitude algorithm for deinterleaving stochastic pulse train sources
IEEE transactions on signal processing, vol. 46, no. 5, pp. 1344–1350, 1998.
 - [17] J. Liu, H. Meng, and X. Wang
A new pulse deinterleaving algorithm based on multiple hypothesis tracking
In *2009 International Radar Conference "Surveillance for a Safer World"(RADAR 2009)*. IEEE, 2009, pp. 1–4.
 - [18] J. Liu, H. Meng, Y. Liu, and X. Wang
Deinterleaving pulse trains in unconventional circumstances using multiple hypothesis tracking algorithm
Signal Processing, vol. 90, no. 8, pp. 2581–2593, 2010.
 - [19] M. Xie, C. Zhao, X. Han, Z. Wang, and D. Hu
Separation of interleaved pulse stream based on directed acyclic graphs
IEEE Signal Processing Letters, vol. 30, pp. 613–617, 2023.
 - [20] W. Cheng, Q. Zhang, J. Dong, C. Wang, X. Liu, and G. Fang
An enhanced algorithm for deinterleaving mixed radar signals
IEEE Transactions on Aerospace and Electronic Systems, vol. 57, no. 6, pp. 3927–3940, 2021.
 - [21] Z.-M. Liu and S. Y. Philip
Classification, denoising, and deinterleaving of pulse streams with recurrent neural networks
IEEE Transactions on Aerospace and Electronic Systems, vol. 55, no. 4, pp. 1624–1639, 2018.
 - [22] M. Zhu, S. Wang, and Y. Li
Model-based representation and deinterleaving of mixed radar pulse sequences with neural machine translation network
IEEE Transactions on Aerospace and Electronic Systems, vol. 58, no. 3, pp. 1733–1752, 2021.
 - [23] W. Chao, S. Liting, L. Zhangmeng, and H. Zhitao
A radar signal deinterleaving method based on semantic segmentation with neural network
IEEE Transactions on Signal Processing, vol. 70, pp. 5806–5821, 2022.
 - [24] Q. Guo, L. Teng, X. Wu, L. Qi, and W. Song
Deinterleaving of radar pulse based on implicit feature
Journal of Systems Engineering and Electronics, vol. 34, no. 6, pp. 1537–1549, 2023.
 - [25] E. P. Kao
An introduction to stochastic processes. Courier Dover Publications, 2019, pp. 51–56.



Jinshuo Hou received the B.E. degree in Automation from University of Science and Technology Beijing, Beijing, China, in 2023.

He is currently working toward the Ph.D. degree in communication and information system with the University of Chinese Academy of Sciences, Beijing, China, and the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China.

His research interests include synthetic aperture radar signal processing and radar reconnaissance signal processing.



Qunying Zhang received the Ph.D. degree in signal and information processing from the Beijing Institute of Technology, Beijing, China, in 1999.

Between 1999 and 2018, she was a Professor with the Institute of Electronics, Chinese Academy of Sciences (CAS), Beijing, China. Since 2019, she has been with the Aerospace Information Research Institute, CAS.

Her research interests include new system of UWB imaging theory and method, UWB radar signal processing and imaging technology.



Peng Zhang received the Ph.D. degree from Space Engineering University, Beijing, China, in 2020.

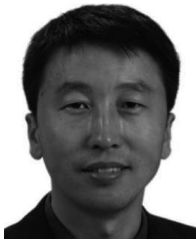
Since 2020, he has been with Beijing Institute of Tracking and Telecommunications Technology.

His current research interests include high resolution imaging method, signal processing.



Wei Lu received the B.S. and M.S. degrees in electronics science and technology from Xi'an Jiaotong University, Xi'an, China, in 2007 and 2010, respectively, and the Ph.D. degree in electromagnetic (EM) field and microwave technology from the University of Chinese Academy of Sciences, Beijing, China, in 2021.

His research interests include ultra-wideband antenna theory and application, ultra-wideband radar system, radar countermeasure, etc.



Xiaojun Liu (Member, IEEE) received the B.S. and M.S. degrees in electronics science and technology from the North University of China, Taiyuan, China, in 1995 and 1998, respectively, and the Ph.D. degree in electrical engineering from the Chinese Academy of Sciences (CAS), Beijing, China, in 2001.

Since 2001, he has been with the Key Laboratory of Electromagnetic Radiation and Sensing Technology, CAS.

His research interests include the application of radars to the remote sensing of the polar ice sheets, ionosphere, and land.



Guangyou Fang received the B.S. degree in electrical engineering from Hunan University, Changsha, China, in 1984, and the M.S. and Ph.D. degrees in electrical engineering from Xi'an Jiaotong University, Xi'an, China, in 1990 and 1996, respectively.

From 1990 to 1999, he was an Engineer, an Associate Professor, and a Professor with the China Research Institute of Radiowave Propagation, Xinxiang, China. From 2000 to 2001,

he was a Visiting Scholar with the University of Trieste, Trieste, Italy, and with the International Center for Science and High Technology, United Nations Industrial Development Organization, Trieste, Italy. From 2001 to 2003, he was a Special Foreign Research Fellow of the Japan Society for the Promotion of Science, working with Prof. M. Sato, Tohoku University, Sendai, Japan. Since 2004, he has been a Professor with the Institute of Electronics, Chinese Academy of Sciences, Beijing, China, and also the Director of the Key Laboratory of Electromagnetic Radiation and Sensing Technology.

His research interests include ultrawideband radars, ground-penetrating radar signal processing and identification methods, terahertz imaging technologies, and computational electromagnetics.